

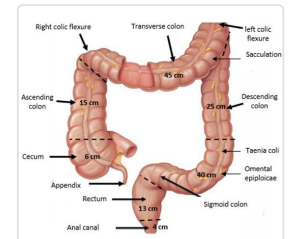
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which parts of the large intestine are intraperitoneal and have a mesentery?
2. What is the commonest position of the appendix, and which is the most dangerous?
3. Give three differences between the anal canal above and below the pectinate line.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Large intestine — overview

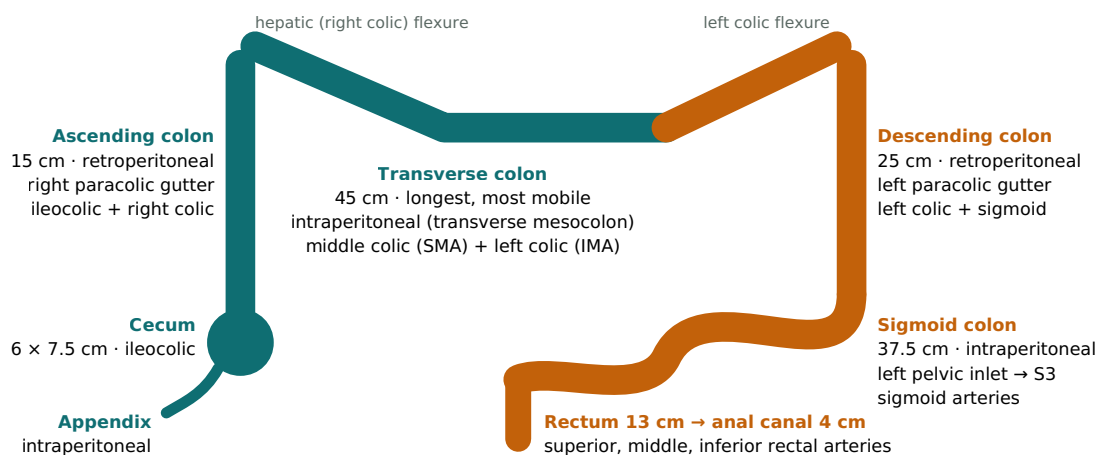
Length	~ 1.5 m : cecum in the right iliac fossa → anus in the perineum
Parts	Cecum and appendix · colon (ascending, transverse, descending, sigmoid) · rectum · anal canal
Peritoneum	All parts are retroperitoneal and fixed except the appendix, transverse colon and sigmoid colon , which are intraperitoneal and have mesenteries



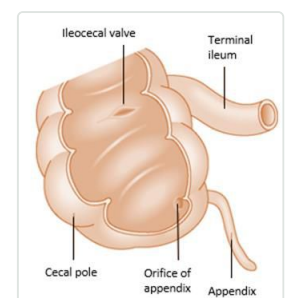
Parts of the large intestine with their lengths.

THE COLON FRAME — length, peritoneum and artery for each part (anterior view)

■ SMA territory ■ IMA territory Intraperitoneal (with mesentery): appendix, transverse, sigmoid. The rest: retroperitoneal, fixed.

**2 Cecum**

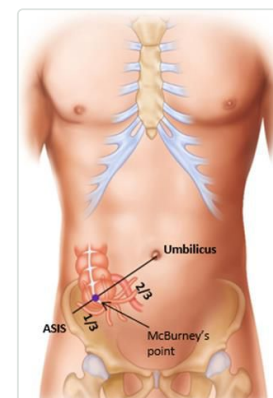
What	Dilated blind sac in the right iliac fossa (L. caecum = blind)
Size	6 cm long × 7.5 cm wide (wider than long) · the highest-pressure area of the colon
Ileal opening	Posteromedial; ileoceleal valve + sphincter control flow in and stop reflux into the ileum
Anterior	Small-intestine coils, greater omentum, anterior abdominal wall
Posterior	Right psoas major and iliacus, femoral nerve , lateral cutaneous nerve of the thigh, right genitofemoral nerve, right gonadal vessels, retrocecal recess (often holds the appendix)
Vessels · lymph	Ileocolic artery (SMA) · veins → SMV → portal · lymph → superior mesenteric pre-aortic nodes
Clinical	Surgeon's reference in intestinal obstruction: distended cecum = large-bowel obstruction ; empty cecum = small-bowel obstruction



Interior of the cecum:
ileoceleal valve and orifice of the appendix.

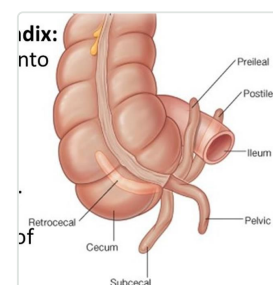
3 Appendix

What	Worm-like diverticulum from the posteromedial wall of the cecum, ~2 cm below the ileocecal junction
Length	2–20 cm (average 9 cm)
Base	Fixed part; surface marking = McBurney's point · all three taeniae coli converge at the base → the surgeon's guide to finding it
Body · tip	Body = narrow tube · tip = least vascular blind end, variable in position



McBurney's point on the anterior abdominal wall.

Position of the tip	Frequency	Note
Retrocecal / retrocolic	65.28%	Commonest
Pelvic	31.01%	Second commonest
Paracolic	2%	
Subcecal	2%	
Pre-ileal · post-ileal	1% · 0.4%	Passes up and medially, tip toward the spleen · pre-ileal is the most dangerous : infection spreads into the general peritoneal cavity



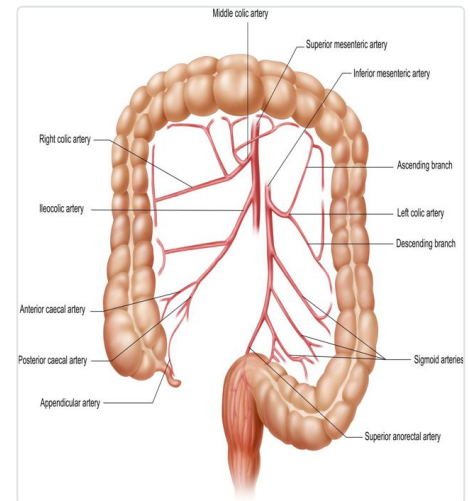
Positions of the appendix tip.

Appendicitis	Common surgical emergency · early pain is referred to the umbilical region (shared T10 segment) · maximum tenderness at McBurney's point
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4 The colon, part by part

	Ascending	Transverse	Descending	Sigmoid
Length	15 cm (6 in)	45 cm (18 in) : longest, most mobile	25 cm	37.5 cm (15 in)
Extent	Cecum → under right lobe of liver → hepatic (right colic) flexure	Between the two flexures; has the transverse mesocolon	Left colic flexure → pelvic brim	Left pelvic inlet → S3 , where it becomes the rectum; S-shaped
Position	Right paracolic gutter	Intraperitoneal	Left paracolic gutter	Intraperitoneal
Arteries	Ileocolic + right colic (SMA)	Middle colic (SMA) + left colic (IMA)	Left colic + sigmoid (IMA)	Sigmoid branches (IMA)
Veins	SMV	SMV and IMV	IMV	IMV → portal
Lymph	Superior mesenteric nodes	Proximal 2/3 → colic → superior mesenteric; distal 1/3 → colic → inferior mesenteric	Colic + inferior mesenteric nodes	Nodes along sigmoid arteries → inferior mesenteric

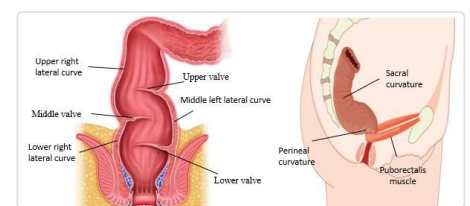
Colostomy	Sigmoid is the usual site in rectal carcinoma : its mobility lets a loop be brought out
Volvulus	Extreme mobility → the sigmoid can rotate around its mesentery
Diverticula	Common in the sigmoid · in diverticulitis or ulcerative colitis it may adhere to neighbours and form internal fistulas
Endoscopy	Close to the anus, so its mucosa can be examined directly



Colic branches of the SMA and IMA.

5 Rectum

Size	~ 13 cm long, 4 cm wide · lower part dilates as the rectal ampulla
Extent	In front of S3 → tip of the coccyx · posterior part of the lesser pelvis
Anterior	Male: bladder , separated by the rectovesical pouch · Female: uterus , separated by the pouch of Douglas* · these pouches limit spread of infection
No colonic features	No taeniae coli , sacculations or appendices epiploicae
Curvatures	Five : three lateral, two antero-posterior



Curvatures of the rectum.

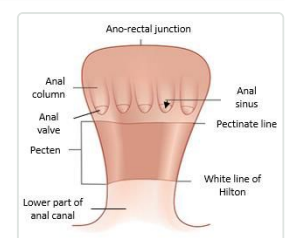
*The slide text calls this pouch “utero-vesical (Douglas' pouch)”. The pouch of Douglas is the rectouterine pouch; the utero-vesical pouch lies between uterus and bladder.

	Upper rectum	Lower rectum
Arteries	Superior, middle and inferior rectal arteries	
Veins	Superior rectal vein → IMV → portal	Middle and inferior rectal veins → internal iliac and internal pudendal veins (systemic)
Lymph	Pararectal nodes → inferior mesenteric nodes	Along the middle rectal artery → internal iliac nodes

Digital rectal exam	Gloved, lubricated finger · feels lumps, tumours, haemorrhoids, rectal carcinoma, prostate , seminal vesicles, ampulla of ductus deferens, bladder, uterus , cervix , ovaries , anorectal abscesses, polyps
Rectal cancer spread	Lymphatics · liver (superior rectal vein is a portal tributary) · posterior → sacral plexus → sciatica · lateral → ureter · anterior → vagina, uterus, bladder, prostate, seminal vesicles

6 Anal canal

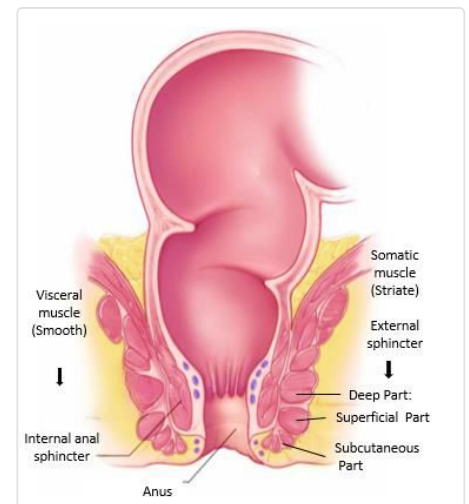
What	Terminal part of the large intestine, 4 cm , in the perineum → anus
Position	Anal triangle, between the fat-filled ischio-rectal fossae , which allow it to expand in defecation
Divided by	The pectinate line : upper $\frac{2}{3}$ (visceral, intestine) · lower $\frac{1}{3}$ (somatic, perineum)



Interior of the anal canal and pectinate line.

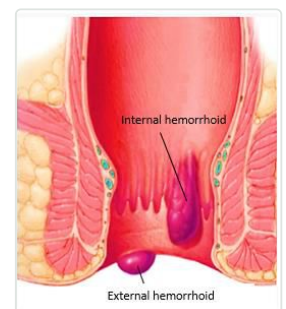
Feature	Upper $\frac{2}{3}$ (above pectinate line)	Lower $\frac{1}{3}$ (below pectinate line)
Development	Endoderm	Ectoderm
Epithelium	Simple columnar	Stratified squamous (non-keratinized → keratinized near the anal verge)
Artery	Superior rectal (IMA)	Inferior rectal (internal pudendal)
Veins	Superior rectal → portal	Inferior rectal → systemic
Lymph	Internal iliac nodes	Superficial inguinal nodes (horizontal group)
Nerves	Autonomic via inferior hypogastric plexus	Somatic: inferior rectal nerve (pudendal)
Haemorrhoids	Internal: painless	External: painful

	Internal sphincter	External sphincter
Control	Involuntary	Voluntary
Made of	Thickened circular muscle coat of the canal	Three parts: deep, superficial, subcutaneous
Extent	Upper $\frac{2}{3}$ of the canal, surrounded by the deep and superficial external parts	Whole length of the canal



Internal and external anal sphincters.

	Internal (true) piles	External (false) piles
Vein	Tributaries of the superior rectal vein	Tributaries of the inferior rectal vein
Site	Above the pectinate line · linked to portal obstruction	Below the pectinate line
Sensation	Painless , sensitive only to stretch → a large one aches	Very painful (inferior rectal nerves: pain, temperature, touch, pressure) · typically do not bleed on straining



Internal and external haemorrhoids.

P Must-remember pearls

- Intraperitoneal with mesentery: **appendix, transverse colon, sigmoid colon**; everything else is retroperitoneal and fixed.
- Appendix: posteromedial cecum, 2 cm below the ileocecal junction · **taeniae converge at its base** · base at McBurney's point.
- Tip positions: **retrocecal 65%** (commonest) > pelvic 31% > paracolic / subcecal 2% > pre-ileal 1% (**most dangerous**).
- Transverse colon: **longest (45 cm), most mobile**; middle colic (SMA) + left colic (IMA); lymph proximal $\frac{2}{3}$ → SM nodes, distal $\frac{1}{3}$ → IM nodes.
- Rectum: 13 cm, S3 → coccyx · **no taeniae, sacculations or appendices epiploicae** · 5 curvatures.
- Rectal cancer → liver via the **superior rectal vein (portal)**; posterior invasion → sacral plexus → **sciatica**.
- Pectinate line: above = endoderm, columnar, superior rectal, portal, internal iliac nodes, autonomic, painless piles · below = ectoderm, squamous, inferior rectal, systemic, **superficial inguinal nodes**, pudendal, painful piles.
- Internal sphincter = involuntary circular muscle · external sphincter = voluntary; deep, superficial, subcutaneous parts.

Q Test yourself

1. A 19-year-old has 12 hours of vague central abdominal pain that has now moved to the right iliac fossa, with maximal tenderness at McBurney's point. Why was the early pain felt around the umbilicus?

- A. The appendix lies behind the umbilicus
- B. Appendix and umbilical skin share the T10 segment
- C. The inflamed appendix irritates the femoral nerve
- D. The pain travels along the ileocolic artery
- E. The appendix is supplied by the pudendal nerve

2. During an appendicectomy the surgeon cannot see the appendix. Which landmark on the cecum should he follow?

- A. The ileocecal valve
- B. The right colic artery
- C. The three taeniae coli, which converge at its base
- D. The hepatic flexure
- E. The retrocecal recess only

3. At laparotomy for intestinal obstruction, the cecum is found collapsed and empty. Where is the obstruction most likely to be?

- A. Sigmoid colon
- B. Descending colon
- C. Rectum
- D. Small intestine
- E. Anal canal

4. A 45-year-old man has a painful, tender swelling at the anal margin, below the pectinate line. Which vein is dilated?

- A. Superior rectal vein
- B. Inferior rectal vein
- C. Inferior mesenteric vein
- D. Middle colic vein
- E. Left gastric vein

5. A squamous carcinoma arises in the lower third of the anal canal. Which nodes should be examined first for spread?

- A. Internal iliac nodes
- B. Inferior mesenteric nodes
- C. Superior mesenteric nodes
- D. Pararectal nodes
- E. Superficial inguinal nodes

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 B** — Visceral pain from the appendix is referred to the umbilical region because both share the T10 segment.
- 2 C** — All three taeniae coli converge at the base of the appendix, so tracing them leads to it.
- 3 D** — An empty cecum means the obstruction is in the small intestine; a distended cecum means it is in the large intestine.
- 4 B** — External haemorrhoids are dilated tributaries of the inferior rectal vein below the pectinate line, and they are painful.
- 5 E** — The lower anal canal drains to the superficial inguinal nodes; the upper part drains to the internal iliac nodes.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy LG 3 slides, 2026-2027
Study alongside the lecture, not instead of it.